Girolamo Fracastoro was one of the first to believe that fossilized shells, such as this fossil of the Archimedes species, had once been animals.

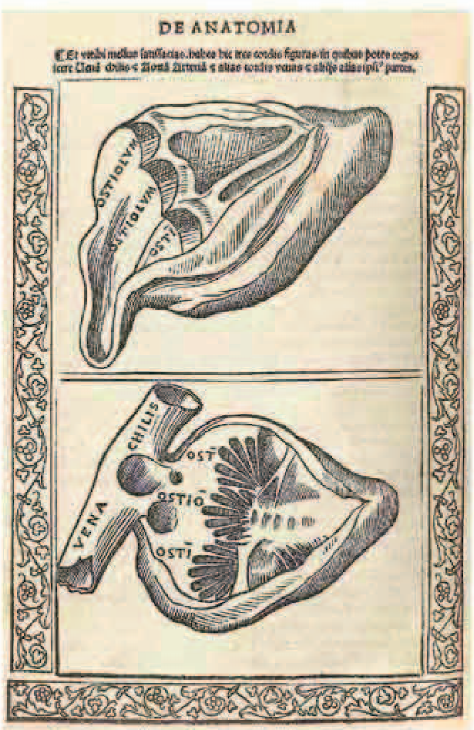
> **" TO KEEP ALIVE THE MEMORY OF OLD KINGDOMS AND EVENTS AND… MAKE KNOWN TO COMING GENERATIONS OUR TIME. "**

Imperial charter describing Gerardus Mercator's terrestrial globe, *c.*1535

body (microcosm) must be in balance with nature (macrocosm). His interest in distilling chemicals led him to use apparently noxious substances such as sulfuric acid (which he employed against gout), mercury, and arsenic as medicines. Some time before 1529 he began to use a **pain reliever he called laudanum**.

In 1533, Flemish cartographer **Gemma Frisius** (1508–55) gave the **first full description of the method of triangulation**, by which a large area could be surveyed from an accurately measured base line. He went on in 1547 to suggest a new way of calculating longitude, by using a portable clock set to the time of the point of departure that could be compared with a clock showing the time at the point of arrival. The imprecision of clocks, however, rendered the technique of limited practical value.

In 1521 Italian surgeon **Berengario da Carpi** (*c.*1460–1530), who lectured on anatomy at the University of Bologna, wrote about the importance of the anatomy of things that can be observed, which included the **use of dissection of human corpses**. He used this as the basis for his *Anatomia Carpi* (*The Anatomy of Carpi*), which was the first anatomical work to use printed figures to illustrate the text.



Attention to detail
Drawings from Anatomia Carpi *show the veins leading into the heart. These examples show how accurately da Carpi derived his drawings from his program of human corpse dissection.*

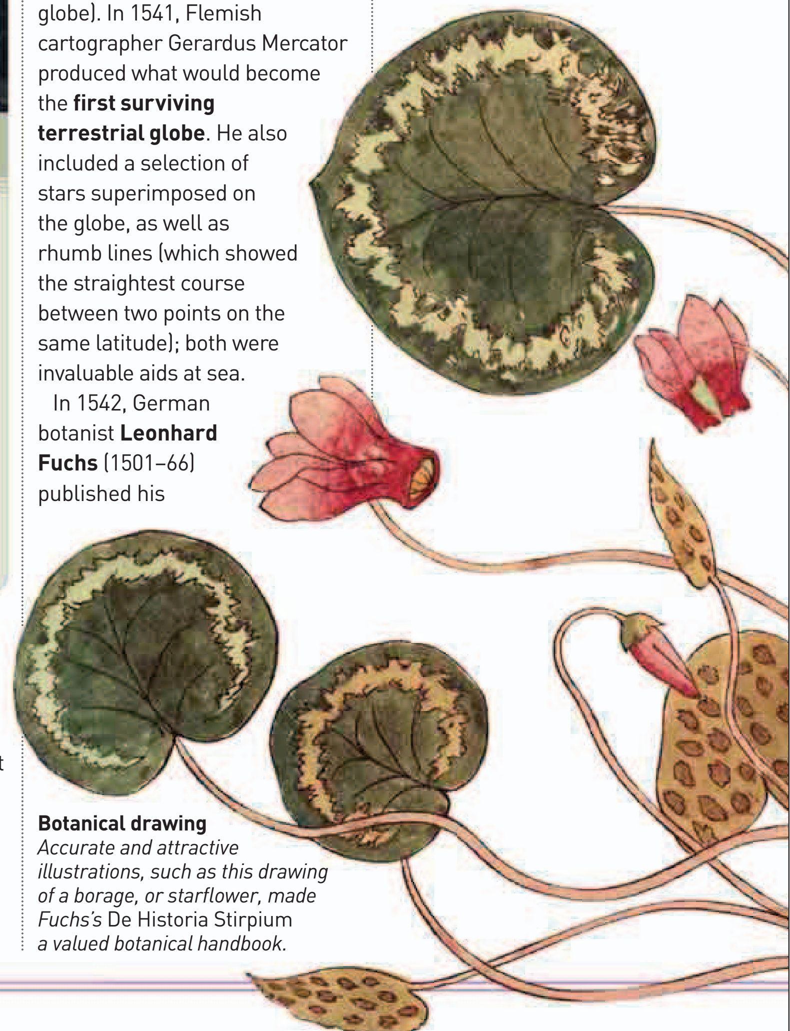
GERARDUS MERCATOR
(1512–94)

Born in Flanders, Gerardus Mercator embarked on a career making mathematical instruments. He began producing maps in 1537 and published his first world map in 1538. In 1569, he compiled another map of the world, this time using a projection that showed constant lines of course as straight lines, which came to bear his name.

AN IMPETUS TO BOTANY IN THE RENAISSANCE had been provided by the desire to illustrate plants found in classical authors' texts, such as those of Roman naturalist Pliny, and the possibilities provided by printed illustrations. Between 1530 and 1536, German Carthusian monk **Otto Brunfels** (*c.*1488–1534) published his *Herbarum Vivae Iconis* (*Living Pictures of Herbs*) with 260 woodcuts of plants, whose accurate detail set an exacting **standard for botanical drawing**.

In 1530, Gemma Frisius had compiled a manual explaining how to construct a globe showing Earth's geography (a terrestrial globe). In 1541, Flemish cartographer Gerardus Mercator produced what would become the **first surviving terrestrial globe**. He also included a selection of stars superimposed on the globe, as well as rhumb lines (which showed the straightest course between two points on the same latitude); both were invaluable aids at sea.

In 1542, German botanist **Leonhard Fuchs** (1501–66) published his *De Historia Stirpium* (*The History of Plants*), in which he **described around 550 plants** (mainly medicinal ones), providing their names and therapeutic virtues. Its drawings were so clear that it became the **first botanical work to be widely used by laymen**.



Botanical drawing
Accurate and attractive illustrations, such as this drawing of a borage, or starflower, made Fuchs's De Historia Stirpium *a valued botanical handbook.*

1527 Paracelsus devises new **classification for chemical substances**

1527 German mathematician Peter Apian produces first work in Europe to describe **Pascal's Triangle**

c.1529 Paracelsus uses **laudanum** as a pain-reliever

1530–36 Otto Brunfels publishes important **botanical handbook**

1533 Peter Apain publishes first **sine tables**

1535 Da Carpi publishes first **anatomical textbook** with printed figures

1537 Italian mathematician Niccolò Tartaglia issues his *Nova Scientia*, first major modern **work on ballistics**

1539 German botanist Jerome Bock (Tragus) classifies plants into **herbs, grasses and trees, or shrubs**

1541 Mercator produces his first **celestial globe**

1542 Leonhard Fuchs's *De Historia Stirpium* accurately **describes 550 plants**